

COURSE PLAN

UPES/Course Plan/Theory & lab/Ver 11/ Effective July 2026

Name of School

Name of the Faculty	Prateek Raj Gautam
Designation	Assistant Professor
Cluster	CSO
Program	B.Tech
Course	Backend Development
Course Code	CSFS3008P
AI-integrated Course	Yes
No. of credits	5
Semester	5
Session	2026-27
Academic Year	2026-27

Signature of Faculty Member

Name: _____

Designation: _____

Date: _____

Signature of Cluster Head

Name: _____

Designation: _____

Date: _____

COURSE PLAN

Prerequisite	Frontend Development - CSFS2004	
Credit	5	
Lecture	Tutorial	Practical
4	0	1

A. The expected Program Outcome are:

PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

B. Expected Program specific Outcome are:

PSO1	Perform system and application programming using computer system concepts, the principles of data structures, algorithm development, problem-solving, and optimization techniques.
PSO2	Apply software development and project management methodologies, integrating principles from both front-end and back-end development, and effectively utilize contemporary tools and technologies.
PSO3	Exhibit a commitment to ethical practices, societal responsibilities, and continuous learning, contributing to the advancement of technology and addressing challenges in diverse computing domains.

C. The expected Course Outcomes are:

THEORY and LAB	
CO 1	Understand Back End Development Fundamentals
CO 2	Create and build web pages and applications.
CO 3	Utilize frameworks and APIs.
CO 4	Apply security best practices and conduct testing.
CO 5	Demonstrate effective team collaboration skills.

D. CO-PO Relationship Matrix

Indicate the relationships by 1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	-	-	-	-	3	-	-	-	1	-	-	-	-	-	-
CO 2	-	-	-	-	2	-	-	-	-	-	-	-	-	-	-
CO 3	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	2	-	-	-	-	-	-	-	-	-	-
CO5	-	-	2	-	2	-	-	-	2	-	-	2	-	-	-
Average	-	-	2	-	2	-	-	-	1.5	-	-	2	-	-	-

E. Course Outcomes assessment plan:

a) Mapping of Course Outcomes with assessment tools

Components Course Outcomes	Assignment	Test/Quiz	AI enabled assessment (semester end project/ test/ quiz)	Mid Semester	End Semester	Any other
CO 1	☐	☒	☐	☒	☒	☐
CO 2	☒	☐	☐	☒	☒	☐
CO 3	☐	☒	☐	☐	☒	☐
CO 4	☒	☐	☐	☐	☒	☐
CO 5	☐	☐	☒	☐	☒	☐
CO 6	☐	☐	☐	☐	☐	☐
CO 7	☐	☐	☐	☐	☐	☐

b) Assessment plan and Analysis: THEORY

	Assessment Tools	Planned	Actual	Highest (Score)	Lowest (Score)
CIA	Quiz 1	Before mid			
	Quiz 2	After mid			
	Assignment 1	Before mid			
	Assignment 2	After mid			
Mid Term Exam		As per SRE			
End Term Exam		As per SRE			

c) Assessment plan and Analysis: Lab

	Assessment Tools	Planned	Actual	Highest (Score)	Lowest (Score)
CIA	Viva- Voce	Before mid sem			
	Lab Records	Before end sem			

F. Course Syllabus Template:

Syllabus Template for Theory

Unit Number	Content
Unit Number	Content
Unit I	Introduction to Backend Development - Server-Side Programming: Overview of backend development and its importance, Client-server architecture and communication protocols, Introduction to HTTP and RESTful APIs, Introduction to backend languages and frameworks (e.g., Python, Node.js, Django, Flask), Introduction to server-side programming languages (e.g., Python, JavaScript), Basics of handling HTTP requests and responses, Managing server-side sessions and cookies, Introduction to server-side rendering and template engines.
Unit II	Data Management, API Development and Frameworks - Introduction to databases (relational and NoSQL), Designing database schemas and models, Performing CRUD operations on databases, Query optimization and indexing, Designing and implementing RESTful APIs, Authentication and authorization mechanisms, Handling API requests and responses, API versioning and documentation.
Unit III	Frameworks and Performance Optimization - Introduction to popular backend frameworks (e.g., Express.js, Django, Ruby on Rails), Working with middleware and routing, Integrating third-party APIs and services, Testing and debugging backend applications, Understanding security vulnerabilities and best practices, Implementing secure authentication and authorization mechanisms, Caching and performance optimization techniques, Error handling and logging.
Unit IV	Deployment and DevOps - Overview of deployment environments (e.g., local, cloud, containers), Monitoring and scaling backend applications, Basics of containerization (e.g., Docker), Continuous integration and deployment (CI/CD), Managing infrastructure configurations, Setting up CI/CD pipelines, Building and packaging applications, Automated testing and code quality checks, Continuous Deployment and Release Management, Strategies for releasing software updates, Managing deployment environments, Release orchestration and rollback strategies.
Unit V	Advanced Topics - Real-time communication with Web-Sockets, Microservices architecture and design patterns: Overview of microservices architecture and its benefits, Service discovery and service registration, Implementing communication between microservices (REST, messaging), Serverless computing: Understanding serverless architecture and its advantages, Building serverless backend systems using platforms like AWS Lambda or Google Cloud Functions, Event-driven serverless architecture and integration with other services, Function-as-a-Service (FaaS), Performance profiling and optimization strategies.

Syllabus Template for lab

Experiment Number	Content
1	Create a web page with all possible elements of HTML5
2	Create a web page with all types of Cascading style sheets
3	Create a Responsive Web page with HTML and CSS
4	Create Responsive web page with Bootstrap
5	Programs to demonstrate JavaScript array, object and functions
6	Client-Side Scripts for Form Validation using JavaScript
7	Programs to familiarise ES6 concepts
8	Programs to demonstrate DOM and event handling
9	Programs using AJAX with HTML, XML and JSON data
10	Programs to familiarize jQuery (2 programs)
11	Create a website with HTML, CSS and JavaScript (implement Ajax)
12	Programs to familiarize Server-Side Scripting using Node JS
13	Programs using Mongo DB database with Node JS

COURSE PLAN DELIVERY

UNIT-I

	Session Plan	Actual Delivery			
Lecture	Topics to be Covered	Lecture	Date	Topics Covered	CO Covered
1	Overview of Backend Development: Importance and Role in Web Development				CO1
2	Client-Server Architecture: Understanding the Model and Communication Protocols				CO1
3	Introduction to HTTP Protocol: Request-Response Cycle, Methods, Status Codes				CO1
4	RESTful APIs: Principles, Design, and Best Practices				CO1
5	Introduction to Backend Languages: Python and JavaScript Overview				CO1
6	Introduction to Backend Frameworks: Django, Flask, Express.js, Node.js				CO1
7	Handling HTTP Requests: GET, POST, PUT, DELETE Methods				CO1, CO2
8	Handling HTTP Responses: Status Codes, Response Headers, Body				CO1, CO2
9	Server-Side Sessions: Managing User Sessions and State				CO1, CO2
10	Cookies: Working with Cookies for Client-Side Data Storage				CO1, CO2
11	Server-Side Rendering: Introduction to Template Engines (EJS, Jinja2)				CO1, CO2

COURSE PLAN DELIVERY

UNIT-II

	Session Plan	Actual Delivery			
Lecture	Topics to be Covered	Lecture	Date	Topics Covered	CO Covered
13	Introduction to Databases: Relational vs NoSQL Databases				CO1, CO2
14	Database Schema Design: ER Diagrams, Normalization, Relationships				CO2
15	Data Modeling: Designing Models for Applications				CO2
16	CRUD Operations: Create, Read, Update, Delete in Databases				CO2
17	Query Optimization: Indexing and Performance Tuning				CO2
18	Designing RESTful APIs: Endpoint Planning and Resource Modeling				CO3
19	Implementing RESTful APIs: Building API Endpoints				CO3
20	Authentication Mechanisms: Basic Auth, Token-based Auth, JWT				CO3
21	Authorization Mechanisms: Role-based Access Control, Permissions				CO3
22	Handling API Requests and Responses: Validation, Serialization				CO3
23	API Versioning and Documentation: Swagger/OpenAPI, Postman				CO3
24	Assignment-I Discussion and Solution Review				CO2

COURSE PLAN DELIVERY

UNIT-III

	Session Plan	Actual Delivery			
Lecture	Topics to be Covered	Lecture	Date	Topics Covered	CO Covered
25	Introduction to Express.js Framework: Setup and Basic Structure				CO3
26	Django Framework: Overview, Architecture, and Setup				CO3
27	Working with Middleware: Role and Implementation in Frameworks				CO3
28	Routing in Backend Frameworks: Route Parameters, Query Strings				CO3
29	Integrating Third-Party APIs and Services				CO3
30	Testing Backend Applications: Unit Testing, Integration Testing				CO4
31	Debugging Techniques: Logging, Debugging Tools, Error Tracking				CO4
32	Security Vulnerabilities: OWASP Top 10, Common Threats				CO4
33	Secure Authentication and Authorization Implementation				CO4
34	Caching Techniques: Redis, Memcached, Browser Caching				CO4
35	Performance Optimization: Code Optimization, Database Optimization				CO4
36	Quiz-II (Unit-III)				CO3

COURSE PLAN DELIVERY

UNIT-IV

	Session Plan	Actual Delivery			
re	Topics to be Covered	Lecture	Date	Topics Covered	CO Covered
37	Overview of Deployment Environments: Local, Cloud (AWS, Azure), VPS				CO4, CO5
38	Containerization Basics: Introduction to Docker and Containers				CO4
39	Working with Docker: Dockerfiles, Images, Containers				CO4
40	Monitoring Backend Applications: Logging, Performance Monitoring				CO4
41	Scaling Backend Applications: Horizontal and Vertical Scaling				CO4
42	Continuous Integration (CI): Principles and Implementation				CO4, CO5
43	Continuous Deployment (CD): Deployment Pipelines and Automation				CO4, CO5
44	Building and Packaging Applications: Build Tools, Dependency Management				CO4
45	Automated Testing in CI/CD: Integration with Pipelines				CO4
46	Managing Infrastructure Configurations: Environment Variables, Secrets				CO4
47	Release Strategies: Blue-Green, Canary, Rolling Updates				CO4
48	Assignment-II Discussion and Solution Review				CO4

COURSE PLAN DELIVERY

UNIT-V

	Session Plan	Actual Delivery			
re	Topics to be Covered	Lecture	Date	Topics Covered	CO Covered
49	Real-Time Communication: Introduction to WebSockets				CO3, CO5
50	Implementing WebSockets: <u>Socket.io</u> , WS Libraries				CO3
51	Microservices Architecture: Overview and Benefits				CO5
52	Service Discovery and Service Registration: Consul, Eureka				CO5
53	Communication Between Microservices: REST, Messaging (RabbitMQ, Kafka)				CO5
54	Serverless Architecture: Understanding Functions-as-a-Service (FaaS)				CO5
55	Building Serverless Backend Systems: AWS Lambda, Google Cloud Functions				CO5
56	Event-Driven Serverless Architecture and Integration with Services				CO5
57	Performance Profiling: Identifying Bottlenecks and Optimization				CO4
58	Mini Project: Problem Statement Selection and Architecture Design				CO1-CO5
59	Project Progress Review, Deployment Strategy and Doubt Clearing				CO1-CO5
60	Mini Project Demonstration, Presentation, and Course Wrap-up				CO1-CO5

COURSE PLAN DELIVERY FOR LAB

	Session Plan	Actual Delivery			
Session	Experiment to be Covered	Session	Date	Experiment Covered	CO Covered
1	Create a web page with all possible elements of HTML5				CO2
2	Create a web page with all types of Cascading Style Sheets				CO2
3	Create a Responsive Web page with HTML and CSS				CO2
4	Create Responsive web page with Bootstrap				CO2
5	Programs to demonstrate JavaScript array, object and functions				CO2
6	Client-Side Scripts for Form Validation using JavaScript				CO2
7	Programs to familiarise ES6 concepts				CO2
8	Programs to demonstrate DOM and event handling				CO2
9	Programs using AJAX with HTML, XML and JSON data				CO2
10	Programs to familiarize jQuery (2 programs)				CO2
11	Create a website with HTML, CSS and JavaScript (implement Ajax)				CO2, CO3
12	Programs to familiarize Server-Side Scripting using Node JS				CO3

13	Programs using Mongo DB database with Node JS				
14	Integrated Lab Project - Backend Application with Database				
15	Lab Record Submission and Viva-Voce				

PERIODIC MONITORING- THEORY

Actual date of completion and remarks, if any

Components		From	To	From	To	From	To
Duration (Mention from and to dates)							
Percentage of Syllabus covered							
Lectures	Planned						
	Taken						
Tutorials	Planned						
	Taken						
Test/quizzes	Planned						
	Taken						
	CO's Addressed						
	CO's Achieved						
Assignments	Planned						
	Taken						
	COs Addressed						
	Cos						

Observations(If any)_____

Signature of Faculty

Signature of Head

Date

Date

PERIODIC MONITORING- LAB

Actual date of completion and remarks, if any

Components		From	To	From	To	From	To
Duration (Mention from and to dates)							
Percentage of Experiment covered							
Experiment	Planned						
	Taken						
Viva-Voice/Test/quizzes	Planned						
	Taken						
	CO's Addressed						
	CO's Achieved						
Any other(please specify)	Planned						
	Taken						
	COs Addressed						
	Cos						

Observations(If any)_____

Signature of Head

Date[illegible]

Signature of Faculty

Signature of Head

G. Target

Target	60
Level-1	60
Level-2	70
Level-3	80

H. Method of Evaluation*

UG	PG
Quizzes/Tests, Assignments, AI enabled assessment – (PBL/ Assignment/ etc.) (50%)	Quizzes/Tests, Assignments, seminar, AI enabled assessment – (PBL/ Assignment/ etc.) (50%)
Mid Examination (20%)	End semester (50%)
End examination (30%)	

*It can be revised as per the assessment scheme of the respective School/Course

I. Passing Criteria

Scale	PG	UG
Out of 10point scale	SGPA – “6.00” in each semester CGPA – “6.00” Min. Individual Course Grade – “C” Course Grade Point – “4.0”	SGPA – “5.0” in each semester CGPA – “5.0” Min. Individual Course Grade – “C” Course Grade Point – “4.0”

*for PG, passing marks are 40/100 in a paper (Composite)

*for UG, passing marks are 35/100 in a paper (Composite)

J. References:

Textbooks Web resources	1. David Herron, "Node.js Web Development", 5th Edition, Packt Publication, 2020.
	2. DT Editorial Services, "HTML5 Black Book", 2nd Edition, Dreamtech Publications, 2016.
	3. Sam Newman, "Building Microservices", 2nd Edition, O'Reilly Publication, 2021.
	https://upessocs.github.io/#dir=/Lectures/Backend%20Development/&file=list.txt
Journals Reference books	Journal of Web Engineering, IEEE Transactions on Software Engineering

MOOCs, Online courses AI Tools	1. Ethan Brown, "Web Development with Node and Express", O'Reilly Media, 2019.
	2. Brad Traversy, "JavaScript from Beginner to Professional", 2021.
	NPTEL: Web Development, Coursera: Full Stack Web Development, Udemy: Node.js, Express, MongoDB
	GitHub Copilot, ChatGPT (for learning assistance), AI-powered code reviewers

SUGGESTIONS FOR FACULTY

- Faculty should keep track of the students with low attendance and counsel them regularly.
- The course coordinator will arrange to communicate the short attendance (as per UPES policy) cases to the students and their parents monthly.
- Topics covered in each class should be recorded in the table of RECORD OF CLASS TEACHING (Suggested Format).
- Internal assessment marks should be communicated to the students twice in a semester.
- The file will be audited by respective IQAC members for theory as well as for lab as per schedule.
- The faculty is required to maintain these files for a period of at least three years.
- This register should be handed over to the head of department, whenever the faculty member goes on long leave or leaves the Colleges/University.
- For labs, continuous evaluation format (break-up given in the guidelines for result preparation in the same file) should be followed.
- The department should monitor the actual execution of the components of continuous lab evaluation regularly.
- Instructor should maintain records of experiments conducted by the students in the lab weekly.
- Instructors should promote students for self-study and to make concept diary, due weightage in the internal should be given under faculty assessment for the same.
- Course outcome assessment: To assess the fulfilment of course outcomes two different approaches have been decided. Degree of fulfillment of course outcomes will be assessed in different ways through direct assessment and indirect assessment. In Direct Assessment, it is measured through quizzes, tests, assignments, Mid-term and/or End-term examinations. It is suggested that each examination is designed in such a way that it can address one or two outcomes (depending upon the course completion). Indirect assessment is done through the student survey which needs to be designed by the faculty (sample format is given below) and it shall be conducted towards the end of course completion. The evaluation of the achievement of the Course Outcomes shall be done by analyzing the inputs received through Direct and Indirect Assessments and then corrective actions suggested for further improvement.
- At the completion of the course, course attainment and other documents should be shared with the program coordinator for computation of Program attainment.
- At the completion of the course Faculty members are suggested to share the innovative teaching techniques along with the course plan (format provided by IQAC).
- Faculties are encouraged to share the master/expert classes evidence (as per the event report format)
- Faculties are also encouraged to include MOOCs,,SWAYAM any other online content and share the evidence of MOOCs courses /online courses referred (as per the event report format).
- Faculties are encouraged to share the evidence related to interventions or initiatives focusing the unique/slow and Fast Learners along with Course Completion files.
- Faculty members are encouraged to adopt AI-enabled pedagogical practices and computational tools to promote experiential, interdisciplinary, and outcome-based learning.

- Faculty can design and leverage AI-based assessments at their discretion, in alignment with individual/school/ course requirements.
- Faculties are required to submit a summary report of the AI based assessments towards the end of the semester especially for AI enabled courses

DISCLAIMER FOR STUDENTS

- This course integrates AI-assisted tools and platforms strictly for educational enhancement, including learning support, visualization, simulation, verification, productivity improvement, and conceptual understanding. Students are encouraged to use AI tools responsibly and thoughtfully while maintaining academic integrity, originality of work, and independent analytical thinking.
- Unless explicitly specified otherwise by the faculty, the maximum permissible use of AI-generated content in assignments, projects, presentations, reports, and other internal assessment deliverables **shall be limited to 20%**. Faculty members may revise this limit upward for specific activities or assessments where AI-assisted work is pedagogically relevant.
- All students are required to disclose and acknowledge the use of AI tools wherever applicable.

INDIRECT ASSESSMENT

Sample format for Indirect Assessment of Course outcomes:

NAME:
<i>ENROLLMENT NO:</i>
SAP ID:
COURSE:
PROGRAM:

Please rate the following aspects of course outcomes of -----.

Use the scale 1-3*

Course Outcomes	Statement	1	2	3
CO1	Understand Back End Development Fundamentals			
CO2	Create and build web pages and applications.			
CO3	Utilize frameworks and APIs.			
CO4	Apply security best practices and conduct testing.			
CO5	Demonstrate effective team collaboration skills.			

- 1 → Low level of attainment
- 2 → Moderate level of attainment
- 3 → High level of attainment
- *